

A decomposition obstruction for the Pirkovskii–Piszczyk $k_\infty(\mathbb{N}^2, V)$ example

Candidate substantial partial result for arXiv:2012.08956

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Status. Candidate substantial partial result, likely valid. The source conjecture is proved for topological *algebra* isomorphisms. For arbitrary locally convex-space isomorphisms, a Banach-step localization and a compact tail-corona reduction are proved; that stronger conjecture remains open.

Source conjecture

Pirkovskii and Piszczyk [1] fix $0 < c_j \leq 1$ with $c_j \rightarrow 0$ and define weights on $\mathbb{N}^2$ by

$$v_n(i, j) = \begin{cases} c_j^n, & i < n, \\ 1, & i \geq n. \end{cases}$$

Write

$$\mathcal{E} = k_\infty(\mathbb{N}^2, V) = \text{ind}_n E_n, \quad E_n = \ell_\infty(\mathbb{N}^2, v_n), \quad p_n(x) = \sup_{i,j} |x_{ij}| v_n(i, j).$$

Their Proposition 5.10(v) proves that the analogous k_0 space is not a direct sum of a normed algebra and a contractible Köthe co-echelon algebra. They conjecture the same assertion for $\mathcal{E}$.

all reflexive Fréchet spaces are distinguished, i.e., their strong duals are barrelled. Clearly, each normed space is distinguished, and a direct sum of two distinguished spaces is distinguished. Therefore E' is distinguished.

On the other hand, it is easily seen that the Köthe set $A = (a_n)_{n \in \mathbb{N}}$ on $\mathbb{N}^2$, where $a_n(i, j) = v_n(i, j)^{-1}$, satisfies the conditions of [17] Corollary 27.18]. Hence $\lambda_1(\mathbb{N}^2, A)$ is not distinguished. This is a contradiction since $E' \cong \lambda_1(\mathbb{N}^2, A)$ (see above).

Applying now [1] Corollary 3.5 and Theorem 3.7], we conclude that $k_0(\mathbb{N}^2, V)$ is not complete. $\square$

Proposition 5.10. *Let V be the weight sequence on $\mathbb{N}^2$ given by (14). Then*

- (i) $k_0(\mathbb{N}^2, V)$ and $k_\infty(\mathbb{N}^2, V)$ are topologically amenable Köthe co-echelon algebras;
- (ii) $k_\infty(\mathbb{N}^2, V)$ is unital;
- (iii) $k_0(\mathbb{N}^2, V)$ is not complete;
- (iv) there is no decomposition $\mathbb{N}^2 = S \sqcup T$ such that $k_\infty(S, V_S)$ is a Banach algebra and such that $k_\infty(T, V_T)$ is a contractible algebra;
- (v) the underlying lcs of $k_0(\mathbb{N}^2, V)$ is not isomorphic to a direct sum of a normed algebra and a contractible Köthe co-echelon algebra.

Proof. Properties (i) and (ii) are mentioned in Example 5.6 while (iii) is contained in Lemma 5.9. To prove (iv) and (v), observe that each contractible co-echelon algebra of order $p > 0$ is a Montel space (for $p < \infty$ this follows from Theorem 4.4 while for $p = \infty$ this is [24, Theorem 12]). Also, if a co-echelon algebra of order 0 is contractible, then its completion is a Montel space [24, Theorem 13]. Now (iv) and (v) follow from Lemmas 5.8 and 5.9 respectively. $\square$

We conjecture that (v) holds for $k_\infty(\mathbb{N}^2, V)$ as well.

REFERENCES

- [1] Klaus D. Bierstedt, Reinhold G. Meise, and William H. Summers. Köthe sets and Köthe sequence spaces. In *Functional analysis, holomorphy and approximation theory (Rio de Janeiro, 1980)*, volume 71 of *North-Holland Math. Stud.*, pages 27–91. North-Holland, Amsterdam-New York, 1982.
- [2] Klaus D. Bierstedt. An introduction to locally convex inductive limits. *Functional analysis and its applications (Nice, 1986)*, 35–133, ICPAM Lecture Notes, World Sci. Publishing, Singapore, 1988.

The two images are from source PDF pages 18 and 19. The phrase “underlying lcs” makes the conjecture stronger than an assertion about algebra isomorphisms: a general linear homeomorphism need not preserve coordinate idempotents.

Proof intuition

For an algebra decomposition, the identity of one summand pulls back to an idempotent of the coordinatewise algebra $\mathcal{E}$. Coordinatewise idempotents are exactly characteristic functions, so every algebra decomposition is already one of the coordinate partitions excluded in source Lemma 5.8.

For a merely linear decomposition, replace idempotents by the continuous projection P onto the normed summand. Baire’s theorem forces its Banach range into one fixed weighted ℓ_∞ step. The complementary projection then has Montel range, hence sends every Banach-step unit ball to a relatively compact subset of the inductive limit. Passing to the quotient of bounded row arrays by row- c_0 arrays isolates the remaining obstruction as a compact perturbation of the tail-corona quotient.

The algebra-isomorphism case

Theorem 1. *There is no topological algebra isomorphism*

$$\mathcal{E} \cong B \oplus C$$

where B is a normed algebra and C is a contractible Köthe co-echelon algebra.

Proof. The space $\mathcal{E}$ is complete [2, Theorem 2.3]. The two summands are closed complemented subspaces, so B and C are complete. In particular, B is Banach. A contractible co-echelon algebra of positive order is Montel; for order zero its completion is Montel [3, Theorems 12 and 13]. Thus completeness makes C Montel in every case.

Let $\Phi : \mathcal{E} \rightarrow B \oplus C$ be the alleged algebra isomorphism. Since $\mathcal{E}$ is unital, so is $B \oplus C$, and $e = \Phi^{-1}(1_B, 0)$ is an idempotent. The multiplication of $\mathcal{E}$ is coordinatewise, hence every coordinate of e is zero or one. There is therefore a partition $\mathbb{N}^2 = S \sqcup T$ with $e = \mathbf{1}_S$. Multiplication by e and $1 - e$ gives topological identifications

$$e\mathcal{E} = k_\infty(S, V_S), \quad (1 - e)\mathcal{E} = k_\infty(T, V_T).$$

The restriction of Φ identifies the first space with the Banach space B and the second with the Montel space C . This contradicts Lemma 5.8 of [1], which forbids exactly such a coordinate partition. $\square$

Localization of any hypothetical linear decomposition

The next lemma is useful independently of the algebra structure.

Lemma 2 (Baire localization). *If a Banach space Y is topologically isomorphic to a linear subspace of $\mathcal{E}$, then its image is contained continuously in one step E_N : for some N and $C < \infty$,*

$$p_N(y) \leq C\|y\|_Y \quad (y \in Y).$$

Proof. Regard each p_n as an extended seminorm on $\mathcal{E}$, with value $+\infty$ outside E_n . Its sublevel sets are closed: they are intersections of the closed coordinate inequalities $|x_{ij}|v_n(i, j) \leq r$. Therefore

$$Y = \bigcup_{n, r \in \mathbb{N}} \{y \in Y : p_n(y) \leq r\}$$

is a countable union of closed sets. By Baire's theorem, one of these sets has nonempty interior. Taking differences gives a neighborhood of zero on which p_N is bounded. Homogeneity then yields the displayed estimate on all of Y . $\square$

Proposition 3 (compact-projection reduction). *Suppose, as locally convex spaces, that*

$$\mathcal{E} = B_0 \oplus M_0,$$

where B_0 is normed and M_0 is a contractible Köthe co-echelon algebra. Let P and $Q = I - P$ be the associated projections. Then:

1. B_0 is Banach and M_0 is Montel;
2. for some N , $B_0 \subset E_N$ continuously;
3. B_0 is complemented in every E_m for $m \geq N$;
4. for every $m \geq N$, $Q|_{E_m} : E_m \rightarrow \mathcal{E}$ is compact in the locally convex sense.

Proof. The first assertion follows exactly as in the first paragraph of Theorem 1. Lemma 2 gives (2). For $m \geq N$, $P|_{E_m}$ is a bounded map into the Banach space B_0 and is the identity on B_0 , proving (3). Finally, the unit ball of E_m is bounded in $\mathcal{E}$; its Q -image is bounded in the Montel space M_0 , hence relatively compact in M_0 and therefore in $\mathcal{E}$. $\square$

Tail-corona form of the remaining obstruction

Put

$$X = \ell_\infty(\mathbb{N}; \ell_\infty(\mathbb{N})), \quad X_0 = c_0(\mathbb{N}; \ell_\infty(\mathbb{N})), \quad Z = X/X_0,$$

and let $q : X \rightarrow Z$ be the quotient map. We identify X with E_1 .

Proposition 4 (tail-corona factorization). *There is a continuous linear map $\Pi : \mathcal{E} \rightarrow Z$ whose restriction to $E_1 = X$ is q . Under the hypotheses of Proposition 3, let $B_0 \subset E_N$ and let $R_N : B_0 \rightarrow X$ delete the first $N - 1$ rows. Then*

$$q = qR_N P|_X + K, \quad K : X \rightarrow Z \text{ compact.}$$

Proof. For $x \in E_n$, delete the first $n - 1$ rows. The remaining array is bounded, with X -norm at most $p_n(x)$. If a later step is used, the two bounded representatives differ in finitely many rows and hence by an element of X_0 . This defines Π unambiguously. Its restriction to each E_n is bounded, so the inductive-limit property makes Π continuous; on E_1 it is q .

For $b \in B_0 \subset E_N$, tail deletion gives $\Pi(b) = qR_N(b)$. On X we have

$$q = \Pi(P + Q)|_X = qR_N P|_X + \Pi Q|_X.$$

By Proposition 3, $Q|_X$ maps the unit ball to a relatively compact subset of $\mathcal{E}$. Its composition $K = \Pi Q|_X$ is therefore compact. $\square$

Scope, verification, and novelty

Theorem 1 strictly extends source Proposition 5.10(iv) from coordinate partitions to all topological algebra decompositions. It does not settle the source wording, which concerns arbitrary locally convex-space isomorphisms. Propositions 3 and 4 give a necessary condition for such an isomorphism.

A tempting final step would use the moving-row copy of c_0 and the fact that B_0 is complemented in a Banach space isomorphic to ℓ_∞ . The gap is that $Q|_{E_N}$ is compact only into the inductive-limit topology, not necessarily in the fixed E_N norm. No claim requiring that invalid upgrade is made.

The idempotent computation, every closed-set step in Lemma 2, and independence of the tail representative were checked directly. No computation is used. A bounded novelty search on 11 August 2026 covered the run indexes and exact-phrases/keyword web searches for the conjecture and its decomposition terms. The source and published version were found, but no later resolution. Novelty confidence is moderate because the ingredients are standard, while this application was not located.

Human-review recommendation. Check first the order-zero completeness point and Lemma 2. For an upgrade, test whether the factorization in Proposition 4 is incompatible with a projection extending continuously to every weighted step; do not replace inductive-limit compactness by fixed-step norm compactness.

References

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